



UMS
Utho Marking Solutions

Serialine.

Installation, commissioning and support – everything a technician needs on site.

TECHNICIAN INSTALLATION MANUAL

JET2neo > JET3 > PORT 3000



For Leibinger JET2neo printers

Internal – UMS technicians

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TECHNICIAN MANUAL

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Golden rules on site

- Work through the stages in order. Each ends with a check that must pass before the next.
- Never run a client's real serial file until Test A (chapter 07) has passed.
- Leave the client with the printed **User Manual**, its support page filled in.

01 OVERVIEW Installation at a glance

A standard Serialline installation takes about **2 hours on site**, plus 30 minutes of preparation before the visit.

1 Build package, collect client info UMS office · 30 min Chapter 03	2 Configure printer interface Printer · 15 min Chapter 04	3 Set up the print job Printer · 20 min Chapter 05
4 Install on client PC and test Client PC · 20 min Chapter 06	5 Product and line-speed tests Line · 30 min Chapter 07	6 Operator training and handover Line · 30 min Chapter 08

02 PREPARATION Kit list

Pack this before leaving for site.

- ☐ USB stick with the `Serialline_Package` folder (`Serialline.exe` , `printer_settings.txt`)
- ☐ USB stick copy of `serialline.py` and `requirements.txt` (fallback if the .exe is blocked)
- ☐ Test file `Test5.csv` with 5 serials (`TEST0001` to `TEST0005`)
- ☐ Line-speed test file with 50 serials
- ☐ Ethernet cable, 3 m or longer
- ☐ Printer service/admin password
- ☐ Printed **UMS Serialline – User Manual** for the client
- ☐ This manual, with the commissioning sheet (chapter 07)
- ☐ Test products (at least 60) agreed with the client

03

PREPARATION

Before the visit

Collect from the client

Client / site	
Printer model and serial number	
Printer firmware version	
Printer IP address on the client network	
Client PC (laptop/PC, Windows version)	
Network: shared network, or direct cable?	
Serial format and maximum length (example)	
Batch size (serials per file)	
Line speed (products per minute)	
Client contact and IT contact	

Build the Serialine package

On a UMS PC with Python installed:

- 1 Put `serialine.py`, `printer_settings.txt` and `build_exe.bat` in one folder.
- 2 Double-click `build_exe.bat` and wait until it shows **Done** (1–3 minutes).
- 3 Check the new folder `Serialine_Package` contains `Serialine.exe` and `printer_settings.txt`.
- 4 Edit `printer_settings.txt` with the client's printer IP, then copy `Serialine_Package` to the USB stick.

```
printer_settings.txt

ip=192.168.1.100
port=3000
```

04 PRINTER Printer configuration

Serialine talks to the printer through the Leibinger interface protocol. The printer silently ignores anything else, so every setting below must match. Log in at service/admin level first.

✓	Menu	Setting	Required value
☐	Settings / Basic settings / IP address	IP and subnet	Client network address, e.g. 192.168.1.100 / 255.255.255.0
☐	Extra / Interface settings / Communication interface	Interface	Ethernet
☐	Extra / Interface settings / Communication interface	Protocol	JET3
☐	Extra / Interface settings / Communication interface	Port	3000
☐	Extra / Database setting	Activate database	OFF

Restart the printer if it asks. Tested on JET2neo firmware V75.0.11.2, interface protocol V1.9.9.

05

PRINTER Print job setup

The job needs one text field that takes its text from the mailing queue. Without it, the printer accepts the serials but prints nothing from them.

- 1 Stop printing and open the client's job in the job editor (or create a new job).
- 2 Add or select the text object for the serial.
- 3 Set its text source to **Extern Text** in **mailing mode** (may be called Mailing, Database or Mail field), **field number 1**.
- 4 Set the **length** to at least the longest serial. Recorded length: _____
- 5 Set font, position and orientation as the client requires. Other fixed text, dates or logos in the job are fine.
- 6 Job parameters: **PrintGo source = External** (photo-eye), edge as wired; PrintGo repetition = 0.
- 7 Set the print delay so the serial lands in the right place on the product.
- 8 **Save the job, then load it again.**

CHECK

The connection test (chapter 06) must show a line like

*EXTTXT [12 (X) 1 0 0 0 0 0] . The **third value must be 1** (mailing field 1); the first value is the length. A 0 in third place means Extern Text mode – fix step 3.

06

CLIENT PC

Client PC setup

- 1 **Copy.** Copy `Serialine_Package` to the client PC, e.g. `C:\Serialine` or the Desktop, and rename the folder to `Serialine`.
- 2 **Network.** Connect the PC to the printer's network. In Command Prompt run `ping <printer IP>`. Replies = OK.
- 3 **No replies?**
 - Direct cable: Settings → Network & internet → Ethernet → IP assignment → Edit → Manual → IPv4. Use an address in the printer's range, e.g. `192.168.1.50`, subnet `255.255.255.0`, no gateway.
 - Client network: ask client IT for the printer's reachable IP and put it in `printer_settings.txt`.
- 4 **Power.** Settings → System → Power → Screen and sleep: sleep when plugged in = **Never**.
- 5 **Antivirus / SmartScreen.** `Serialine.exe` is not code-signed. On "Windows protected your PC" click More info → Run anyway. If antivirus quarantines it, ask client IT to allow it, or use the fallback (chapter 10).
- 6 **Connection test.** Double-click `Serialine.exe`, type **T**, press ENTER. All six pass criteria below must show.
- 7 **Shortcut.** Right-click `Serialine.exe` → Show more options → Send to → Desktop (create shortcut). Rename it "Serialine".

```
Serialine – connection test (pass)

--- 192.168.1.100:3000 ---
Version      : [... 'V75.0.11.2', 'JET2neo', ...]      ← 1 printer found
Machine state: Ready for print start, error 0         ← 2 no error
Mailing FIFO : depth 256, entries 0, last printed 0   ← 3 mailing available
Loaded job    : \\FFSDISK\\JOBS\\CLIENT_JOB.job       ← 4 correct job
Extern-text fields in the job:
  *EXTTXT [ 12 (X) 1 0 0 0 0 0]                      ← 5 third value = 1

OK - this port works. Use --port 3000                ← 6
```

07 COMMISSIONING

Commissioning and sign-off

Commissioning passes when every box is ticked. Keep the signed page with the job card.

Test A – 5 products

File `Test5.csv` ; printer ready, printing stopped.

- ☐ Shows `5 serial(s) queued ... (records 1..5)`
- ☐ Products 1–5 get `TEST0001` – `TEST0005` in order
- ☐ `[SUCCESS]` and the printer stops by itself
- ☐ A 6th product after the stop gets **no** print
- ☐ `Test5_log.csv` lists the 5 serials
- ☐ Print position, size and legibility approved

Test B – line speed

50-serial file at normal production speed.

- ☐ All 50 printed in order, no gaps, no duplicates
- ☐ No printer errors during the run
- ☐ Products per minute: _____

Test C – stop and resume

- ☐ Stop mid-batch shows `[STOPPED] ... record X`
- ☐ After ENTER, printing resumes at X+1

Test D – reprint

- ☐ Finished file offers reprint; `YES` reprints; old log kept

AFTER SIGN-OFF

Delete the test files and test logs from the Serialine folder.

UMS TECHNICIAN

Name _____

Signature _____

Date _____

CLIENT REPRESENTATIVE

Name _____

Signature _____

Date _____

08 HANDOVER Handover and operator training

Train every operator who will run batches, and let each one run a test batch themselves.

- ☐ Fill in the Support page of the **User Manual** (technician, install date, printer serial, printer IP) and hand it over
- ☐ Show how to prepare a serial file: column A, from A1, no heading, unique serials
- ☐ Operator starts a batch alone: printer ready → double-click Serialine → choose file → check N → ENTER
- ☐ Show a stop and resume
- ☐ Show where the logs are and explain they must be kept
- ☐ Explain reprints: remove the first-run products first, then **y** and **YES**
- ☐ Explain the blank product that can occur at the end of a batch
- ☐ Agree who backs up the Serialine folder, and where

Operators trained

09 SUPPORT Technical troubleshooting

Start every diagnosis with the connection test: `Serialine.exe`, then T.

Symptom	Cause	Fix
Port 3000: <code>actively refused it</code>	Interface not listening on 3000	Interface = Ethernet, port 3000; restart printer
<code>timed out</code> / ping fails	PC not on the printer's network	Fix IP/subnet or cable (chapter 06, step 3)
Connects, <code>No reply to ?VS</code>	Protocol not JET3, or wrong device	Protocol = JET3; check the IP is the printer
<code>EXTTXT [.. (X) 0 ...]</code>	Field in Extern Text mode	Mailing mode, field 1; save and reload job
No <code>EXTTXT</code> line	Job has no external text field	Add the mailing field (chapter 05)
<code>did not accept the serials into its mailing queue</code>	Database active, no mailing field, or job not reloaded	Activate database OFF; check field; reload job
Error 5043 after the last serial	Mailing queue ran empty (old version)	Use the current <code>Serialine.exe</code>
<code>stopped after record 0</code> right after ENTER	Printer stopped before the first product	Check the printer screen; press ENTER to resume
<code>error/message NNNN</code>	Message pending on the printer	Acknowledge; look up NNNN in the printer's error list
Numbering error at batch start	Firmware expects a sequence starting at 0	Report to UMS development with firmware version
Every trigger prints, no new serial	Mailing field missing; fixed text printing	Check the EXTTXT line; reload the job
No print on a product	Sensor not triggering, or PrintGo internal	Check photo-eye LED; PrintGo source = External
<code>does not match the data file</code>	File edited after printing began, or log name reused	Keep the original file; use a new file name for a new batch
Blank product after the last serial	Product triggered before Serialine stopped the printer	Expected at high speed; remove it

10

REFERENCE

Command line and fallback

Open Command Prompt in the Serialine folder. Without `--file`, Serialine starts in operator mode (file list).

Task	Command
Connection test	<code>Serialine.exe --test-connection</code>
Print a file (log = <code><file>_log.csv</code>)	<code>Serialine.exe --file Batch.xlsx</code>
Print with a chosen log name	<code>Serialine.exe --file Batch.xlsx --log Batch_A_log.csv</code>
Reprint a whole batch	<code>Serialine.exe --file Batch.xlsx --reprint</code>
Reprint records 120–135 only	<code>Serialine.exe --file Batch.xlsx --reprint --from 120 --to 135</code>
Other printer IP or port, once	add <code>--ip 192.168.1.120 --port 3000</code>
Start without pressing ENTER	add <code>--yes</code> (also skips the reprint confirmation – use with care)

FALLBACK

If the .exe is blocked: install Python 3 from python.org for the current user (tick "Add python.exe to PATH"). Copy `serialine.py`, `requirements.txt` and `printer_settings.txt` into the Serialine folder and run `pip install -r requirements.txt` there. Double-click `serialine.py`, or use the commands above with `python serialine.py` in place of `Serialine.exe`.

11

UNDER THE HOOD

How Serialine talks to the printer

Serialine uses the printer's **mailing mode** (Leibinger Interface Protocol v1.9.15, Annex B). Every message is `^@` + command + carriage return, over TCP port 3000.

Command	Purpose in Serialine
<code>?VS</code>	Firmware version (connection test)
<code>?RS</code>	Machine state (5 = ready, 6 = printing) and error number
<code>?SM</code>	Mailing queue: depth, entries, last printed record number
<code>?JB</code> / <code>?JL</code>	Loaded job script / name (checks the EXTTEXT field)
<code>!FF</code>	Clear the mailing queue before loading a batch
<code>=CM<n></code>	Stop after record n (ignored by JET2neo V75.0.11.2)
<code>=MR<n><TAB><serial></code>	Queue serial as record n
<code>!GO</code> / <code>!ST</code>	Start / stop printing

Sequence per batch

- 1 Wait for state 5 (ready); clear the queue (`!FF`).
- 2 Send up to 255 records (`=MR`), numbered 1, 2, 3 ... in file order.
- 3 Operator presses ENTER → `!GO`. Each PrintGo prints the next record; the printer rejects gaps and repeats.
- 4 Every 0.2 s Serialine reads `?SM`, logs newly printed records and tops up the queue in blocks of 50. The queue is a rolling buffer, so batch size is unlimited (tested with 12,000 serials in one run).
- 5 A blank end-marker record follows the last serial so the queue never runs empty (JET2neo error 5043). When the last serial is reported printed, Serialine sends `!ST`.
- 6 If printing stops early, the printer clears its queue; Serialine reloads from the first unprinted record.



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